CivicConnect • Milestone 3 Release Candidate Baseline

Belgium Campus iTversity • SEN381 Software Engineering 381 (NQF Level 8) • Group E

Project Engineering Document (PED) v3.0

CivicConnect: Community Service Request Management Platform

Milestone 3 — Controlled Construction, Integration, Quality & Release Readiness

Academic Institution: Belgium Campus iTversity
Module Code & Name: SEN381 — Software Engineering 381 (NQF Level 8)
Project Name: CivicConnect (Community Service Request Management Platform)
Document Reference Identifier: `DOC-PED-003` (Controlled Release Candidate Baseline)
Baseline Version: 3.0 (Supersedes `DOC-PED-002` v2.0 Baselined on 2026-09-30)
Release Candidate Identifier: `v0.3.0-rc1`
Governing Standard: SEN381 Milestone 3 Brief (`SEN381_Project_Milestone_3.pdf`)
Assessment Allocation: 50 Raw Marks (30 Team Evidence Marks + 20 Individual Defence Marks)

Required Submission Information Block (M3 Brief §5.3)

Submission Field	Official Project Baseline Value
Team Designation & Name	**CivicConnect Group E**
Active Team Members	• **Chris Fourie** (Student ID: **602826**) — Systems Architect & Governance Lead (~~80% Workload)~~ ~~• **Lisa Verson** (Student ID: **602006**) — Lead Requirements & Quality Analyst (~~20% Workload) *(Note: Pandora Greyling [602369] withdrew on 2026-09-29; absorbed under ratified ADR-009).*
GitHub Repository URL	`https://github.com/SEN381-CivicConnect/Project` (Assessed Local Root: `C:\Chris\Studies\SEN\Project`)
M3 Release Candidate Identifier	Tag: `v0.3.0-rc1`
Baseline Cutoff Commit (M2)	Commit: `ffc17cb` (2026-10-01 00:33:49 +0200)
PED Version	**v3.0** (Evidence Index Baseline)
RTM Location / Link	`docs/requirements/Requirements_Traceability_Matrix_RTM_v3.0.md`
CI / Quality Gate Evidence	Workflow: `.github/workflows/ci.yml` (Passing 4 quality gates, 41/41 tests passing)
Staging Environment Instructions	Multi-stage Docker Compose on port `5001` (DB on `5433`): `docs/architecture/Staging_Deployment_and_Rollback_Guide.md`
Date Submitted	3 October 2026

Document Control & Revision History

Version	Date	Primary Authors	Approval Authority	Milestone Scope & Lifecycle Status
1.0	2026-09-09	Chris Fourie, Pandora Greyling, Lisa Verson	Two-Reviewer Policy (`ADR-002`)	**M1 Baseline:** Problem analysis, 14 FRs, 10 NFRs, scope boundaries, FSM, RTM v1.0, Risk Register v1.0, ADR-001–003.
2.0	2026-09-30	Chris Fourie, Lisa Verson	Two-Person Review (`ADR-009`)	**M2 Baseline:** Clean Architecture, Weighted Stack Matrix (`ADR-008`), Strict 3NF DDL (`ADR-006`), Factory/Observer patterns, Web portal, RTM v2.0, Risk v2.0.
3.0	2026-10-03	Chris Fourie, Lisa Verson	Two-Person Review (`ADR-009`)	**M3 Release Candidate Baseline:** Implemented release candidate, dual persistence adapter (`ADR-010`), staging parity (`ADR-011`), 41 automated tests, CI pipeline, Defect Register v3.0, Tech Debt Register v1.0, RTM v3.0, and Release Evidence Summary.

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Section 1: Milestone 3 Overview & Executive Baseline

Milestone 3 marks the transition of CivicConnect from the architectural specifications established in Milestone 2 into a **controlled, verified, and demonstrable release candidate (`v0.3.0-rc1`)**.

Per **SEN381 Milestone 3 Brief §1 & §3**, M3 answers the central engineering question:

> *"Can the team demonstrate, with manageable and credible evidence, that CivicConnect has been constructed, integrated and verified well enough to become a credible release candidate — while clearly identifying what still remains uncertain or incomplete?"*

Implemented Scope Summary

The release candidate integrates all 14 baselined Functional Requirements:

- **Citizen Fault Intake (`FR-001` , `FR-002`):** Polymorphic category validation via Factory Method (`ADR-005`), automated SLA binding (`FR-013`), emergency prefixing, and POPIA privacy toggle.
- **Operational Lifecycle & FSM (`FR-009` , `FR-010` , `FR-011`):** Finite State Machine enforcing allowable state transitions (`SUBMITTED` → `ASSIGNED` → `IN_PROGRESS` → `RESOLVED` → `CLOSED`), optimistic concurrency control versioning (`ADR-006`), and mandatory resolution details.
- **Security & POPIA Defenses (`FR-008` , `NFR-004` , `NFR-005`):** Controller-level RBAC role authorization guards returning HTTP 403 Forbidden for unauthorized modifications, and field-level masking protecting citizen identities from unauthorized staff views.
- **Operational Monitoring & Dashboard (`FR-003` , `FR-006` , `FR-012`):** Interactive web dashboard, real-time ticket triage, and active synthetic health probes (`/health/live` , `/health/ready`).

Section 2: Milestone 2 Baseline Conformance

M2 Architectural Decision	Baselined Intent (PED v2.0)	Implemented State in M3 Release Candidate	Conformance Status	Implementation & Verification Evidence
Clean Layered Monolith	Strict inward dependency flow (`domain` → `application` → `infra` / `pres`).	Implemented cleanly across 4 directories in `code/src/` . Zero cyclic imports.	**100% Conforming**	`code/src/domain/` , `src/application/` , `src/presentation/` , `src/infrastructure/`
Relational Model & 3NF	Strict 3NF schema in PostgreSQL 16 with declarative foreign keys and checks.	DDL migration script executed in Docker PostgreSQL with constraints (`chk_role_code` , `chk_priority_code`).	**100% Conforming**	`code/database/migrations/V1__initial_schema.sql`
Optimistic Concurrency (OCC)	Integer `version` column checked on ticket update (`ADR-006`).	Enforced in domain `ServiceRequest.ts` and repository SQL (`WHERE version = $4`); HTTP 409 mapped.	**100% Conforming**	`OptimisticConcurrency.test.ts` `api.test.ts` `DEF-001` fix
Factory Method Pattern	Specialized creators per request category (`ADR-005`).	`ServiceRequestFactoryRegistry` routes 9 intake rules with category-specific validation invariants.	**100% Conforming**	`CategoryFactories.ts` `IntakeDecisionTable.test.ts` (10 tests)
Observer Pattern	Decouple ticket updates from notification sinks (`ADR-004`).	`DomainEventDispatcher` dispatches immutable events to `AuditLoggingObserver` and `NotificationDispatchObserver` .	**100% Conforming**	`DomainEventDispatcher.ts` `ObserverPattern.test.ts`
Environment Parity	Local container replication matching cloud PaaS (`DEC-005`).	Multi-stage `Dockerfile` and `docker-compose.staging.yml` mirror production Linux runtime.	**100% Conforming**	`Dockerfile` `docker-compose.staging.yml`

Section 3: Controlled Architectural Changes

In accordance with **SEN381 M3 Brief §7 & §22**, all significant deviations from Milestone 2 were formally justified, evaluated, and documented in Architecture Decision Records rather than occurring silently:

1. `ADR-009` **: Governance Adjustment for Two-Person Team Operation (Accepted 2026-09-29)**
   - *Problem:* Departure of Pandora Greyling (602369) made the mandatory 2-reviewer PR rule (`ADR-002`) mathematically impossible.
   - *Justified Change:* Amended branch protection to require **100% peer review approval from the remaining partner (1 mandatory review)** combined with automated CI pipeline quality gates.
   - *Consequence:* Development proceeded without administrative deadlocks while maintaining 100% peer review coverage.
2. `ADR-010` **: Dual Persistence Architecture (Accepted 2026-10-02)**
   - *Problem:* Forcing PostgreSQL container startup inside GitHub Actions runners added ~60s latency per PR and threatened free-tier minute exhaustion.
   - *Justified Change:* Introduced dual persistence via Clean Architecture interface: fast `InMemoryServiceRequestRepository` for CI (<2s test runs) and `PostgresServiceRequestRepository` for Docker staging, toggled via `USE_POSTGRES` .
   - *Consequence:* Fast feedback loop in CI runners without compromising production PostgreSQL 16 schema integrity.
3. `ADR-011` **: Simulated Staging Parity, Secret Isolation & Rollback Strategy (Accepted 2026-10-02)**
   - *Problem:* Satisfying M3 staging and disaster recovery requirements without incurring cloud hosting costs.
   - *Justified Change:* Created simulated staging via Docker Compose on isolated ports (`5001/5433`) with environment-specific secrets (`.env.staging`) and a declarative reverse migration script (`V1__rollback_initial_schema.sql`).
   - *Consequence:* Full compliance with M3 Brief §14 at $0.00 cost.

Section 4: Construction & Configuration Management

- **Repository Strategy:** Strict trunk-based feature branching. Direct pushes to `main` are blocked. All changes merge via formal Pull Requests utilizing `.github/pull_request_template.md`.
- **Configuration & Secrets Management:**
  - Zero credentials or API keys exist in source code or Git history.
  - Local development uses `.env` defaults; staging uses `.env.staging` (strictly ignored by `.gitignore`).
  - Automated security scans in CI (`npm audit --omit=dev --audit-level=high`) verify zero high-risk vulnerabilities.
- **Component Monorepo Structure:**

```
Project/
├── .github/workflows/ci.yml       # Multi-gate automated CI pipeline
├── code/
│   ├── database/migrations/       # Strict 3NF DDL & Rollback migrations
│   ├── public/                    # Interactive web portal (zero-build dashboard)
│   ├── src/                       # Clean Architecture backend codebase
│   │   ├── domain/                # Entities, FSM invariants, Factory Method, Events
│   │   ├── application/           # Use cases, DTO mappers, Observers
│   │   ├── infrastructure/        # In-Memory & PostgreSQL Repository Adapters
│   │   └── presentation/          # Express REST controllers and health routes
│   ├── tests/                     # 41 automated tests (Unit, BB, API, E2E, Load)
│   ├── Dockerfile                 # Multi-stage production container build
│   └── docker-compose.staging.yml # Isolated simulated staging environment
└── docs/                          # Living engineering registers, RTM v3.0, and PED
```

Section 5: CI & Automated Verification Quality Gates

CivicConnect enforces a four-stage automated quality pipeline configured in `.github/workflows/ci.yml`. Every pull request and push to `main` must pass all gates; any single failure blocks merge progression:

```
flowchart LR
    G1["Gate 1: Restore Dependencies<br><code>npm ci</code>"] --> G2["Gate 2: Static Analysis<br><code>tsc --noEmit</code> & build"]
    G2 --> G3["Gate 3: Automated Verification<br><code>vitest run --coverage</code>"]
    G3 --> G4["Gate 4: Security Audit<br><code>npm audit (high+)</code>"]

    style G1 fill:#1e3a8a,stroke:#3b82f6,color:#fff
    style G2 fill:#1e3a8a,stroke:#3b82f6,color:#fff
    style G3 fill:#065f46,stroke:#10b981,color:#fff
    style G4 fill:#065f46,stroke:#10b981,color:#fff
```

- **Gate 1 (Dependencies):** Verifies deterministic dependency tree using `npm ci`.
- **Gate 2 (Static Analysis):** Enforces strict TypeScript compilation (`"strict": true` in `tsconfig.json`) ensuring type safety, correct interfaces, and zero compilation errors.
- **Gate 3 (Automated Verification):** Executes 41 automated tests in Vitest with V8 coverage collection.
- **Gate 4 (Security Vulnerability Gate):** Scans production dependencies for high or critical CVE advisories.

Section 6: Quality Strategy & Risk-Based Verification

Testing effort was allocated based on quantitative risk exposure identified in `Project_Risk_Register_v3.0.md`:

1. **Optimistic Concurrency Control (`RSK-002`, Exposure 16 → 3):** Concurrent technician claiming creates data loss. Verified across 3 independent test tiers: entity unit tests (`OptimisticConcurrency.test.ts`), API integration tests (`api.test.ts`), and end-to-end user journeys (`lifecycleJourney.test.ts`).
2. **Polymorphic Intake Rules (`RSK-004`, Exposure 12 → 2):** Divergent category validation rules risk regression bugs. Verified via a 9-rule Decision Table matrix (`IntakeDecisionTable.test.ts`) and boundary value analysis (`InputBoundaryValidation.test.ts`).
3. **POPIA Citizen Privacy (`RSK-006`, Exposure 15 → 3):** Citizen phone/email exposure violates South African data privacy laws. Verified via unit DTO tests, API role tests, and E2E whistleblower journeys ensuring PII is masked for field staff.

Section 7: Black-Box, Integration & System Validation Evidence

CivicConnect delivers **41 automated test cases** across 10 test suites (substantially exceeding the M3 minimum of 15 cases):

```
Test Suite Summary:
✓ tests/unit/entities/OptimisticConcurrency.test.ts  (2 tests)
✓ tests/unit/factories/ServiceRequestFactory.test.ts  (4 tests)
✓ tests/unit/entities/ServiceRequestFSM.test.ts  (4 tests)
✓ tests/unit/blackbox/IntakeDecisionTable.test.ts  (10 tests)
✓ tests/unit/blackbox/InputBoundaryValidation.test.ts  (10 tests)
✓ tests/unit/events/ObserverPattern.test.ts  (1 test)
✓ tests/unit/dtos/AnonymizationMask.test.ts  (2 tests)
✓ tests/integration/api.test.ts  (5 tests)
✓ tests/e2e/lifecycleJourney.test.ts  (2 tests)
✓ tests/performance/load.test.ts  (1 test)
Total: 41 passed across 10 files (Execution duration: 1.30s)
```

Representative Verification Evidence Records (M3 Brief §11)

Evidence Record 1: Boundary Value Analysis on IT Support Description (`BB-BVA-01`)

- **Test Basis:** `FR-001`, `AC-001.2`, `RSK-004`
- **Why Selected:** Verifies input boundary validation preventing vague, un-actionable IT fault tickets.
- **Technique:** Boundary Value Analysis (Threshold $N = 10$ characters; inputs evaluated at $N - 1 = 9$, $N = 10$, $N + 1 = 11$).
- **Input / Precondition:** Input string `'123456789'` (9 characters) sent to IT creator factory.
- **Expected Result:** Throws domain validation exception: `"Validation Failure (IT_SUPPORT): Description must specify device or connectivity details (min 10 characters)."`.
- **Actual Result:** Throws expected validation error matching regex pattern.
- **Status: PASS** (Evidence: `code/tests/unit/blackbox/InputBoundaryValidation.test.ts#L21-L35`).
- **Interpretation:** Proves boundary enforcement rejects sub-threshold inputs while accepting valid descriptions at $N = 10$.

Evidence Record 2: Decision Table Routing Matrix for Facilities (`BB-DEC-01`)

- **Test Basis:** `ADR-005`, `FR-001`, `FR-002`, Rule R1 in Decision Table.
- **Why Selected:** Verifies multi-condition business logic routing physical building faults to Facilities with High priority.
- **Technique:** Decision Table Testing.
- **Input / Precondition:** Category `'FAC_FAULT'`, Title `'Pipe leak'`, Location `'Main Building, Room 102'`.
- **Expected Result:** Ticket created with `departmentId=1` (Facilities), `priority=HIGH`, `slaResolutionHours=24`.
- **Actual Result:** Matches expected department and SLA parameters exactly.
- **Status: PASS** (Evidence: `code/tests/unit/blackbox/IntakeDecisionTable.test.ts#L30-L46`).
- **Interpretation:** Validates automated routing invariants across category taxonomy.

Evidence Record 3: Concurrency Conflict Handling on Ticket Assignment (`API-INT-03`)

- **Test Basis:** `FR-009`, `ADR-006`, `RSK-002`, `DEF-001`
- **Why Selected:** Prevents lost updates when two supervisors simultaneously dispatch technicians.
- **Technique:** API Integration Testing with Supertest against Express REST API.
- **Input / Precondition:** Ticket created at `version=1`. First supervisor assigns technician with `expectedVersion: 1` (increments to version 2). Second supervisor sends `expectedVersion: 1`.
- **Expected Result:** Backend detects stale version and responds with `HTTP 409 Conflict` containing `"Concurrency conflict"`.
- **Actual Result:** HTTP 409 Conflict returned; second assignment rejected.
- **Status: PASS** (Evidence: `code/tests/integration/api.test.ts#L43-L71`).
- **Interpretation:** Proves OCC non-blocking concurrency eliminates race conditions without pessimistic DB table locking.

Evidence Record 4: E2E Full Operational Lifecycle Journey (`E2E-J1`)

- **Test Basis:** `FR-001`, `FR-008`, `FR-009`, `FR-010`, `FR-011`, `NFR-006`
- **Why Selected:** Validates complete integrated system flow across all architectural layers.

- **Technique:** End-to-End System Journey Testing.
- **Input / Precondition:** Multi-step journey from citizen submission → supervisor inspection → technician assignment → progress update → resolution with notes → supervisor closure → terminal FSM lock check.
- **Expected Result:** Status transitions strictly sequentially; versions increment from 1 to 5; resolution details enforced; reopening closed ticket rejected with HTTP 400.
- **Actual Result:** All 8 lifecycle assertions pass in sequence.
- **Status: PASS** (Evidence: `code/tests/e2e/lifecycleJourney.test.ts#L22-L144`).
- **Interpretation:** Validates end-to-end business value across the entire integrated application runtime.

Section 8: Performance & Scalability Evaluation (NFR-001)

Per **SEN381 M3 Brief §13**, the team conducted a manageable, justified load test against the primary performance-sensitive operation: **Service Request Intake (`POST /api/v1/requests`)**.

- **Why Performance Matters:** Ticket intake experiences sudden spikes during localized campus incidents (e.g. power outages, water bursts). High latency degrades citizen trust and leads to request duplication.
- **Workload & Environment:** 50 concurrent requests executed within single-node test runner (`code/tests/performance/load.test.ts`).
- **Target Metric (`NFR-001`):** p95 response time ≤ 500ms under 50 concurrent active users; 0% error rate.
- **Empirical Test Results:**
  - Total Requests Executed: **50**
  - Successful Requests (HTTP 201): **50 (100%)**
  - Failed / Dropped Requests: **0 (0.0% error rate)**
  - Minimum Latency: **0.8 ms**
  - Median Latency (p50): **2.1 ms**
  - 95th Percentile Latency (p95): **5.4 ms** (Substantially beneath 500ms target)
  - Process Memory RSS: **~42 MB** (Well within 512MB free-tier limits)
- **Honest Engineering Interpretation & Limitations:**
  - *What the test proves:* Proves the in-memory Clean Monolith pipeline, validation rules, event dispatching, and DTO serialisation handle high concurrent bursts without memory leaks or race conditions.
  - *What the test does NOT prove:* Does not prove geographic network latency across 3G/4G municipal cellular networks, distributed PostgreSQL database write lock contention under tens of thousands of concurrent citizens, or disk I/O bottlenecks under massive binary attachment uploads.

Section 9: Defects & Quality Interpretation

All anomalies identified during verification were formally logged in `Defect_Register_v3.0.md`:

- `DEF-001` **(High — Concurrency 500 Error):** Controller threw unhandled error on OCC conflict. Corrected by mapping domain `ConcurrencyConflictError` to `HTTP 409 Conflict`. Verified in `api.test.ts`.
- `DEF-002` **(High — Whitespace Notes Bypass):** FSM allowed empty whitespace notes when resolving tickets. Corrected by adding trimmed length check in `ServiceRequest.ts`. Verified in `InputBoundaryValidation.test.ts`.
- `DEF-003` **(Medium — Header Role Fallback):** Dashboard role selector header `x-user-role` was ignored. Corrected by adding `getCallerRole` in `RequestController.ts`. Verified in `api.test.ts`.

Quality Metric Analysis

- **Automated Test Count:** 41 automated tests (100% passing across 10 files).
- **Code Coverage (V8 Provider): 84.03% Statement Coverage**, **83.33% Branch Coverage**, **89.02% Function Coverage**.
- **Coverage Interpretation:** High coverage across core business invariants (`domain/factories` 100%, `domain/events` 97.2%, `domain/entities` 95.8%, `application/use-cases` 96%). Coverage evidence demonstrates that core FSM invariants and business rules are protected against regression, but the team recognizes that code coverage does not guarantee the complete absence of distributed timing defects.

Section 10: Release Readiness Review & Residual Risk Statement

10.1 Release Evidence Summary Table (M3 Brief §15)

Evidence Dimension	Evaluation Result	Engineering Interpretation	Release Action / Governance Link
Build & CI Pipeline	**PASS**	Dependencies restore cleanly; strict TypeScript compiles with 0 errors.	Automated CI blocks failing PRs on `main`.
Automated Regression	**PASS**	41/41 automated tests pass across unit, black-box, API, and E2E suites.	Core business rules and FSM invariants protected.
API & Integration	**PASS**	REST endpoints conform to OpenAPI 3.0; HTTP 403, 404, 409 handled correctly.	Verified in `api.test.ts` with Supertest.
System & E2E Journeys	**PASS**	Full citizen lifecycle journey and POPIA privacy journey pass end-to-end.	Verified in `lifecycleJourney.test.ts`.
Static & Security Audit	**PASS**	Strict TypeScript static analysis passes; `npm audit` reveals 0 high/critical CVEs.	Moderate `uuid` advisory accepted as `DEBT-004`.
Performance Benchmark	**PASS**	50 concurrent intake operations achieve p95 = 5.4ms vs 500ms target (`NFR-001`).	Verified in `load.test.ts` ; 0% error rate.
Known Defects	**CLOSED**	3/3 identified defects (`DEF-001` – `DEF-003`) resolved with automated regression tests.	Tracked in `Defect_Register_v3.0.md`.

10.2 Final Release Classification

In accordance with **Milestone 3 Brief §15**, the team classifies CivicConnect as:

CONDITIONALLY READY

- **Engineering Rationale:** All mandatory functional requirements (`FR-001` to `FR-014`), security authorization guards (`NFR-004`), POPIA masking (`NFR-005`), concurrency controls (`NFR-007`), and automated CI quality gates pass without failure. Zero release-blocking defects remain.
- **Conditions & Limitations:** Release readiness is conditional on:
  1. Operating within student simulated staging environments rather than multi-zone distributed municipal cloud infrastructure.
  2. Executing automated CI runs via the high-performance in-memory repository harness (`ADR-010`).
  3. External SMS/Email notifications operating via decoupled in-memory observers rather than production third-party gateway contracts (`DEBT-003`).

10.3 Residual Risk Statement Table (M3 Brief §16)

Residual Risk	Why It Remains	Current Engineering Mitigation	Recommended Next Action / Owner
Production-Scale Concurrency Uncertainty (`RSK-012`)	Student workstation staging cannot represent city-wide concurrency with thousands of simultaneous citizen app connections.	Executed 50-concurrency load benchmark (`load.test.ts`); Clean Monolith memory footprint verified < 50MB.	Conduct distributed load testing on cloud staging before live municipal rollout. *Owner: Chris Fourie*
In-Memory CI Concurrency Discrepancy (`DEBT-001`)	In-memory Map repository used in CI to ensure < 2s feedback and avoid free-tier runner exhaustion.	Real PostgreSQL 16 adapter implemented (`PostgresServiceRequestRepository.ts`) and verified in Docker staging.	Integrate Testcontainers for ephemeral PostgreSQL instances in M4. *Owner: Chris Fourie*
External Gateway Sandbox Uncertainty (`RSK-013`)	Third-party SMS/Email gateway sandboxes (Twilio/SendGrid) may experience network timeouts in production.	In-memory Observer Pattern (`ADR-004`) isolates domain transactions from gateway latency.	Operationalize cron worker polling `outbox_messages` table in M4. *Owner: Lisa Verson*

Appendix E: Client Change Request (CR-001) Formal Impact Analysis

In strict accordance with **SEN381 Master Project Brief §14 & Appendix E**, all change requests received after baseline sign-off undergo a structured impact analysis across all engineering dimensions before implementation:

E.1 Change Request Metadata

- **Change Request Identifier:** `CR-001`
- **Title:** Automated Urgent Priority Escalation for Safety Hazards & POPIA Citizen Data Masking Toggle
- **Originating Stakeholder:** Municipal Client Steering Committee / Lead Academic Assessor
- **Date Submitted:** 2026-10-01
- **Evaluation Lead:** Lisa Verson (Lead Requirements Analyst) & Chris Fourie (Systems Architect)
- **Change Governance Disposition: APPROVED & FULLY IMPLEMENTED** (Ratified under `ADR-009`)

E.2 10-Dimension Impact Analysis Matrix (Master Brief Appendix E Standard)

#	Impact Dimension	Detailed Engineering Evaluation & Project Evidence
1	**Requirements & Scope**	Affects `FR-002` (Taxonomy & Routing) and `FR-008` (Inspection & Masking). Extends acceptance criteria to require automated escalation of `SAFETY_HAZARD` or `MAJOR` severity issues to `CRITICAL` / `HIGH` priority, and mandatory `[POPIA PROTECTED]` masking for technician roles.
2	**Architecture & Design**	Fulfills Clean Architecture principles: intake routing encapsulated within `CategoryFactories.ts` (Factory Method pattern) and evaluated via Decision Table logic. Domain boundary remains pristine.
3	**UI & Interaction (SO14)**	Extends citizen intake form on `code/public/index.html` with an explicit "Mask My Personal Information" toggle (`isAnonymizedDisplay`). Technician card renders masked citizen phone/address.
4	**Data & Persistence**	Fully backward-compatible. Supported by existing `service_requests` table and `is_anonymized_display` boolean column. Zero destructive database schema migration required.
5	**API Contracts (SO16)**	Updated `POST /api/v1/requests` payload contract to accept optional `isAnonymizedDisplay` (defaults to `true`). Non-breaking contract expansion.
6	**Security & Privacy (POPIA)**	Strengthens compliance with South Africa's POPIA (Act 4 of 2013). Prevents field contractors from accessing citizen phone numbers and private addresses without supervisor authorization.
7	**Quality & Verification**	Added 10 automated Decision Table black-box tests in `code/tests/unit/blackbox/IntakeDecisionTable.test.ts` (`BB-DEC-01` to `BB-DEC-10`) and 2 DTO masking tests in `code/tests/unit/dtos/AnonymizationMask.test.ts`. All 12 tests pass in CI.
8	**Schedule & Resources**	Implementation completed in ~3 hours under `ADR-009` allocation (Chris Fourie 80%, Lisa Verson 20%). Zero schedule slippage against Milestone 3 target delivery.
9	**Deployment & Staging**	Container environment and secrets unchanged. Verified in local Docker staging (`docker-compose.staging.yml`) on port 5001.
10	**Risk & Technical Debt**	Minimal risk of misrouting; covered by automated decision table assertions. Zero technical debt incurred.

Controlled Appendices & Living Register Links

- **Requirements Traceability Matrix (RTM v3.0):** `docs/requirements/Requirements_Traceability_Matrix_RTM_v3.0.md`
- **Project Risk Register (v3.0):** `docs/risk/Project_Risk_Register_v3.0.md`
- **Defect Register (v3.0):** `docs/risk/Defect_Register_v3.0.md`
- **Technical Debt Register (v1.0):** `docs/architecture/Technical_Debt_Register_v1.0.md`
- **AI Usage Register (v3.0):** `docs/governance/AI_Usage_Register_v3.0.md`
- **Engineering Decision Log (ADR-001 to ADR-011):** `docs/decisions/Engineering_Decision_Log.md`
- **Staging Deployment & Rollback Guide:** `docs/architecture/Staging_Deployment_and_Rollback_Guide.md`
- **Milestone 3 Baseline Sign-Off Gate Record:** `docs/governance/Baseline_Sign_Off_Gate_M3.md`
- **Pull Request PR-008 Peer Review Sign-Off Record:** `docs/governance/Pull_Request_M3_Signoff.md`